

MANIPAL UNIVERSITY JAIPUR

Department of Electrical Engineering
Course Handout

A. Basic Details:

Programme Name:	B. Tech (Electrical & Computer Engg.)
Course Name:	Electrical Machines
Course Code:	ELC2104
LTPC (Lecture Tutorial Practical Credits):	3 1 0 4
Session:	July- Nov. 2026
Class:	III Sem.
Course Coordinator:	Dr. Vinay Gupta
Course Instructor(s):	Dr. Vinay Gupta
Additional Practitioner(s) – if any	N/A

- A. Introduction:** The objective of this course provides a basic understanding of AC & DC machinery fundamentals, constructional features, operational analysis through phasor diagrams, equivalent circuits, determination of performance parameters, testing and applications.

B. Course Outcomes:

CO Statement	CO	Bloom's Level	Target Attainment %	Target Attainment level
Explain the construction, operating principles, equivalent circuits, voltage regulation, and parallel operation of single-phase and three-phase transformers	ELC2104.1	Bloom's Level: Understand	75	3
Describe the construction, operating principles, characteristics, armature reaction, commutation, and starting and speed control methods of DC machines.	ELC2104.2	Bloom's Level: Understand	75	3
Analyze the performance of three-phase induction machines using equivalent circuits, torque-slip characteristics, losses, efficiency, no-load and blocked-rotor tests, and explain starting, speed control, and induction generator operation.	ELC2104.3	Bloom's Level: Application, Synthesis	75	3
Explain the construction, operating principles, synchronization, and two-reaction theory of synchronous machines and analyze their operating characteristics.	ELC2104.4	Bloom's Level: Application, Synthesis	75	3

C. Program Outcomes and Program Specific Outcomes:

- [PO.1]. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
- [PO.2]. **Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
- [PO.3]. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
- [PO.4]. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to
- [PO.5]. provide valid conclusions
- [PO.6]. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations
- [PO.7]. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice
- [PO.8]. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development
- [PO.9]. **Ethics:** Apply ethical principles and commit to professional ethics_and responsibilities and norms of the engineering practices
- [PO.10]. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings
- [PO.11]. **Communication:** Communicate effectively_on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions
- [PO.12]. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments
- [PO.13]. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
- [PO.14]. To solve complex practical problems related to electrical & electronics engineering applications by applying and correlating the knowledge gained from mathematics, basic sciences and other fundamental courses.
- [PO.15]. To design, develop and analyse the prevalent domains of electrical systems for sustainable, reliable, environmental friendly and feasible solutions.
- [PO.16]. Develop, investigate and solve different models of electrical networks using modern engineering tools for variety of real time, industrial and research problems.

D. Assessment Plan:

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Class Work Sessional (CWS): Quiz, MOOC, Assignment.	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

E. Syllabus:

Transformers: Single phase & Three phase transformers – introduction, equivalent circuit, voltage regulation, parallel operation. **DC Machines:** Working principle, construction, types, armature reaction, commutation, characteristics, starting and speed control methods of DC motor. **Induction Machines:** Three-phase induction machine, construction, type and working principle, equivalent circuit, losses and efficiency, torque-slip characteristics, no-load and blocked rotor tests, starting, speed control, induction generator. **Synchronous Machines:** Construction and working principle, synchronization, two-reaction theory.

REFERENCE BOOK

1. D. P. Kothari & I. J. Nagrath, Electric Machines (4e), TMH, 2013.
2. E. H. Langsdorf, Theory of Alternating Current Machinery (2e), TMH, 2004.
3. P. S. Bhimbra, “Electrical Machinery”, Seventh Edition, 1995, Khanna Publishers.
4. I. J. Nagrath, , D. P. Kothari, “Electric Machines”, Third Edition, Tata McGraw-Hill Publishing Company Ltd.
5. A. E. Fitzgerald, Charles Kingsley, Jr. Stephen D. Umans, “Electric Machinery”, Fifth Edition, Tata McGraw-Hill.

F. Lecture Plan:

Lec. No.	Topics	Session Outcome	Corresponding CO	Mode of Delivery	Mode of Assessing the Outcome
1	Introduction	To acquaint students with the outcome-based education (OBE) and Course outcome (CO) and program outcome (PO) assessment process			
2	Introduction to Electrical Machines, Electromechanical Energy Conversion	Explain the classification and applications of electrical machines	CO1	Lecture Based Teaching-Learning	Oral Questions
3	Single-Phase Transformer: Construction & Working Principle	Describe the construction and operation of single-phase transformers	CO1	Technology Based Learning	Quiz
4-5	EMF Equation & Equivalent Circuit	Analyze the equivalent circuit of transformers	CO1	Learning through Problem-Solving	Tutorial Problems
6-7	Voltage Regulation & Efficiency	Calculate voltage regulation and efficiency	CO1	Learning through Problem-Solving	Assignment

Lec. No.	Topics	Session Outcome	Corresponding CO	Mode of Delivery	Mode of Assessing the Outcome
8-9	Parallel Operation of Transformers	Explain the conditions for parallel operation	CO1	Group-Teaching and Learning	Quiz
10	Three-Phase Transformers & Connections	Describe three-phase transformer connections	CO1	Technology Based Learning	Tutorial
11-12	Numerical Problems on Transformers	Solve transformer-related numerical problems	CO1	Learning through Problem-Solving	Assignment
13	DC Machines: Construction & Types	Explain the construction and types of DC machines	CO2	Lecture Based Teaching-Learning	Oral Questions
14-15	Working Principle & EMF Equation	Describe the operating principle of DC machines	CO2	Technology Based Learning	Quiz
16	Armature Reaction	Explain armature reaction and its effects	CO2	Inquiry Based Learning	Tutorial
17-19	Commutation & Characteristics	Describe commutation and characteristics of DC machines	CO2	Peer Teaching	Assignment
20	Starting Methods of DC Motors	Explain different starting methods	CO2	Kinaesthetic Learning	Quiz
21-22	Speed Control Methods	Analyze speed control techniques	CO2	Learning through Problem-Solving	Tutorial
23-24	Numerical Problems on DC Machines	Solve DC machine performance problems	CO2	Learning through Problem-Solving	Assignment
MIDTERM EXAM					
25-26	Three-Phase Induction Machine: Construction	Explain the construction and operation of induction machines	CO3	Technology Based Learning	Oral Questions
27	Rotating Magnetic Field & Slip	Explain slip and rotating magnetic field	CO3	Inquiry Based Learning	Quiz
28-29	Equivalent Circuit	Analyze the equivalent circuit	CO3	Learning through Problem-Solving	Tutorial
30-31	Losses & Efficiency	Calculate losses and efficiency	CO3	Technology Based Learning	Assignment
32-33	Torque-Slip Characteristics	Interpret torque-slip characteristics	CO3	Inquiry Based Learning	Quiz
34-35	No-load & Blocked Rotor Tests	Analyze induction motor performance using tests	CO3	Kinaesthetic Learning	Tutorial

Lec. No.	Topics	Session Outcome	Corresponding CO	Mode of Delivery	Mode of Assessing the Outcome
36-38	Starting & Speed Control	Explain induction motor starting and speed control methods	CO3	Group-Teaching and Learning	Assignment
39	Induction Generator	Describe induction generator operation	CO3	Flipped Classroom	Quiz
40-45	Synchronous Machine: Construction & Synchronization	Explain construction and synchronization	CO4	Lecture-Based Teaching-Learning	Oral Questions
46-48	Two-Reaction Theory, Revision & Problem Solving	Explain two-reaction theory and solve numerical problems	CO4	Inquiry-Based Learning	End-Sem Tutorial / Quiz
END TERM EXAM					

G. Outside class engagement activities

Type of Activity	Difficulty Level	Hours Engaged
Self-study of transformer construction, working principle, EMF equation and equivalent circuit using prescribed textbooks	Low	3
Solving additional numerical problems on transformer equivalent circuit, voltage regulation and efficiency	Low	3
Watching curated video lectures on transformer operation, three-phase connections and DC machine operation	Low	2
Preparation of comparative notes on DC machine types, characteristics, starting and speed-control methods	Low	2
Peer-learning/group discussion on armature reaction, commutation and effects of armature reaction	Moderate	3
Simulation/virtual experimentation of transformer and DC machine characteristics using suitable engineering tools	Moderate	3
Solving numerical problems on induction-machine equivalent circuit, slip, losses, efficiency and torque-slip characteristics	Moderate	3
Analysis of no-load and blocked-rotor test data of a three-phase induction motor	Moderate	2
Case study on industrial applications of induction motors, induction generators and variable-speed drives	High	2
Problem-solving activity on starting or speed control of electrical machines	High	3
Technical seminar/webinar/expert lecture on modern electrical machines, drives or renewable-energy applications	High	2
Self-study and numerical/problem-solving activity on synchronous-machine synchronization and two-reaction theory	High	2
Total		30

H. Learner-centric activities:

Activity
Concept quizzes on transformer fundamentals, DC machines and induction machines
Step-by-step numerical exercises on transformer regulation and efficiency
Video-based learning on rotating magnetic field, slip and machine operation
Comparative study of DC motor starting and speed-control methods
Peer teaching on armature reaction and commutation
Group analysis of torque-slip characteristics
Virtual/simulation-based study of machine characteristics
Analysis of no-load and blocked-rotor test results
Case study of industrial motor selection and speed control
Mini-project on electrical-machine performance or control
Research-based presentation on induction generators in renewable-energy systems
Higher-order problem solving involving synchronous machines and two-reaction theory

I. Course Articulation Matrix:

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES												CORRELATION WITH PROGRAM SPECIFIC OUTCOMES		
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
ELC2104.1	Explain the construction, operating principles, equivalent circuits, voltage regulation, and parallel operation of single-phase and three-phase transformers	3	2	-	-	3	-	-	-	-	-	-	3	3	2	2
ELC2104.2	Describe the construction, operating principles, characteristics, armature reaction, commutation, and starting and speed control methods of DC machines.	3	2	3	3	3	3	-	-	3	3	1	2	3	2	-
ELC2104.3	Analyze the performance of three-phase induction machines using equivalent circuits, torque-slip characteristics, losses, efficiency, no-load and blocked-rotor tests, and explain starting, speed control, and induction generator operation.	3	3	3	3	3	2	-	-	3	3	1	3	3	2	2
ELC2104.4	Explain the construction, operating principles, synchronization, and two-reaction theory of synchronous machines and analyze their operating characteristics.	3	3	2	3	3	2	-	-	3	3	1	3	3	2	2